

- 4 There are nearly 2 000 species of blow fly. They have a worldwide distribution.

Blow flies lay their eggs on dead mammals.

The eggs hatch into larvae that feed on organic matter.

The photograph shows some blow fly larvae feeding on muscle.



(Source: © Minden Pictures / Alamy Stock Photo)

Magnification $\times 1$

Larvae are grown in large numbers in laboratories to test the toxicity of new chemical products.

In a laboratory they are fed on muscle.

A student formed the following hypothesis:

Blow fly larvae will have a higher rate of respiration when feeding on cubes of liver compared with cubes of muscle.

Plan an investigation to find evidence to support or reject this hypothesis.

Your answer should give details under the following headings.

(a) Describe preliminary practical work that you might undertake to ensure your proposed method would provide quantitative results.

(2)

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(b) Devise a detailed method, including how you would control and monitor important variables.

(9)

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(3)

(2)

TOTAL FOR PAPER = 50 MARKS